

Risk	Existing Control	Effectiveness	Gap Identified	Recommended Control
PRODUCT LISTING PAGE (PLP)				
Incorrect price displayed	Automated price sync from pricing DB; periodic batch	Partially effective	No real-time monitoring for sudden price mismatch	Add real-time price validation (API-based), exception alerts
Out-of-stock items shown as available	Inventory API sync every few seconds	Adequate	Occasional lag under high traffic	Implement fail-safe "inventory threshold buffer" and soft OOS checks
Wrong discount/offer applied	Promo engine validation rules	Partially effective	Edge cases where stacking logic fails	Add promo rule regression tests; approval workflow for new offers
PRODUCT DETAILS PAGE (PDP)				
Incorrect product specs/info	Seller self-upload checks + random QC	Partially effective	Manual checks miss details	Introduce automated spec-template validation; seller rating-based QC
Wrong seller information visible	Seller mapping rules + marketplace policy	Adequate	Delays in seller status updates	Add real-time seller status sync + auto-block for non-compliant sellers
ADD TO CART				
Price mismatch between PDP & cart	Cart pulls price from central pricing DB	Partially effective	Rare outdated cache entries	Force real-time price refresh when adding to cart
Coupons applied incorrectly	Promo engine validation at cart	Partially effective	No multi-case validation	Implement coupon misuse detection; automated rule testing suite
CART SUMMARY				
GST breakdown incorrect	GST rules engine + HSN mapping	Partially effective	Manual HSN mismatches possible	Add automated HSN reconciliation; audit trail for tax changes
Incorrect delivery charges	Courier API & rule engine	Partially effective	No fallback if courier API lags	Add geo-based charge fallback & anomaly detection
Incorrect quantity (duplicate taps)	Cart quantity validation	Partially effective	No debounce for fast taps	Add client-side + server-side debounce rule
LOGIN / AUTHENTICATION				
OTP authentication failure	Redundant OTP provider integration	Adequate	Retry limit not optimized	Add secondary fallback (WhatsApp/SMS), OTP delivery monitoring
Account takeover risk	Device binding, rate limiting	Partially effective	Missing adaptive MFA	Add behavioral MFA, login anomaly detection
ADDRESS SELECTION				
Invalid address accepted	Address format validation	Ineffective	Logic only checks format, not real validity	Add third-party address verification (Google Maps API)
Pincode mismatch with delivery availability	Pincode-serviceability mapping	Partially effective	Delayed updates from courier partners	Auto-sync courier serviceability daily + SLA exceptions
DELIVERY SLOT/OPTIONS				
Wrong delivery SLA shown	SLA fetched from courier partner API	Partially effective	API delays cause stale SLA data	Add SLA cache refresh, SLA KPI monitoring, fallback "range-based" SLA
PAYMENT				
Duplicate payments due to retry	Payment gateway retry logic + idempotency keys	Partially effective	Idempotency not enforced across all gateways	Enforce idempotent transaction IDs platform-wide
Wrong EMI interest calculated	EMI rules fetched dynamically	Partially effective	Edge-case errors for non-standard banks	Add EMI rule validation + comparison against RBI-mandated tables
Wallet/Gift card balance misapplied	Wallet service with atomic balance deduction	Partially effective	Race conditions during flash sales	Add balance reservation system + concurrency locks
PAYMENT GATEWAY				
Payment captured but order fails	Payment status reconciliation service	Partially effective	Delayed reconciliation	Introduce real-time webhook reconciliation and auto-order creation
Session timeout during payment	Session expiry timers	Partially effective	Timeout thresholds inconsistent across gateways	Standardize timeouts + introduce "resume payment" feature
ORDER CONFIRMATION				
No order ID generated	Atomic order creation transaction	Partially effective	Edge failures during peak load	Add message-queue-based order creation with retry
Invoice not generated/incorrect	Automated invoice service	Partially effective	HSN/GST mismatches	Implement invoice validation service + GST sandbox testing
BACKEND LOGGING				
Missing transaction logs	Centralized logging + ELK stack	Partially effective	Log retention not uniform	Enforce mandatory log retention policy (90+ days)
No reconciliation between payment report & order DB	Daily reconciliation batch	Ineffective	Delayed error detection	Add real-time reconciliation dashboard + automated mismatch alerts
SELLER / MARKETPLACE RISKS				
Wrong seller assigned	Geographic + pricing rules engine	Partially effective	Rule overrides not monitored	Add seller assignment audit logs + rule change approvals
Unverified/blacklisted seller	Seller KYC + periodic checks	Partially effective	Manual review delays	Automate KYC validity checks + block expired KYCs
Seller manipulates price pre-checkout	Price-lock on add-to-cart	Partially effective	Price-lock expiry too short	Extend price-lock duration + alert for sudden seller price spikes
DELIVERY & OPERATIONS RISKS				
Wrong courier selected	Courier selection engine	Partially effective	Regional mapping outdated	Add weekly partner-master auto-sync
Incorrect surge/delivery fee	Rule engine + location-based fees	Ineffective	Inconsistent fee rules	Add fee simulation testing + geo-pricing audit
ETA not updated due to backend lag	ETA fetched from logistics API	Partially effective	High API latency	Add ETA health-check + fallback ETA estimator
RETURNS, REFUNDS, AND POST-ORDER RISKS				
Refund auto-trigger fails	Refund scheduler job	Partially effective	Failures not escalated	Add refund failure alerts + 24-hour SLA monitoring
Incorrect return policy shown	Policy engine mapped by category	Partially effective	Category misclassification	Validate return policy during SKU upload
Refund issued to wrong method	Refund gateway mapping	Ineffective	Weak validations	Add "refund destination confirmation" + mapping audit
TAX, INVOICE, AND LEGAL RISKS				
Incorrect GST/HSN code	HSN master table	Ineffective	Outdated codes	Weekly automated sync with GSTN master
B2B invoice generated wrongly	Business account flag	Partially effective	Inaccurate profile tagging	Add PAN/GST validation during checkout
Missing KYC for high-value orders	Manual KYC request	Ineffective	No automation	Auto-trigger KYC workflow for high-value cart values
DATA & SECURITY RISKS				
API request tampering → price manipulation	API signature validation	Partially effective	Weak rate limiting	Add HMAC signing + dynamic rate limits
Sensitive payment data without encryption	Encryption at rest + PCI-DSS	Adequate	Occasional plaintext logs	Mask payment fields in logs + enforce PCI logging filters